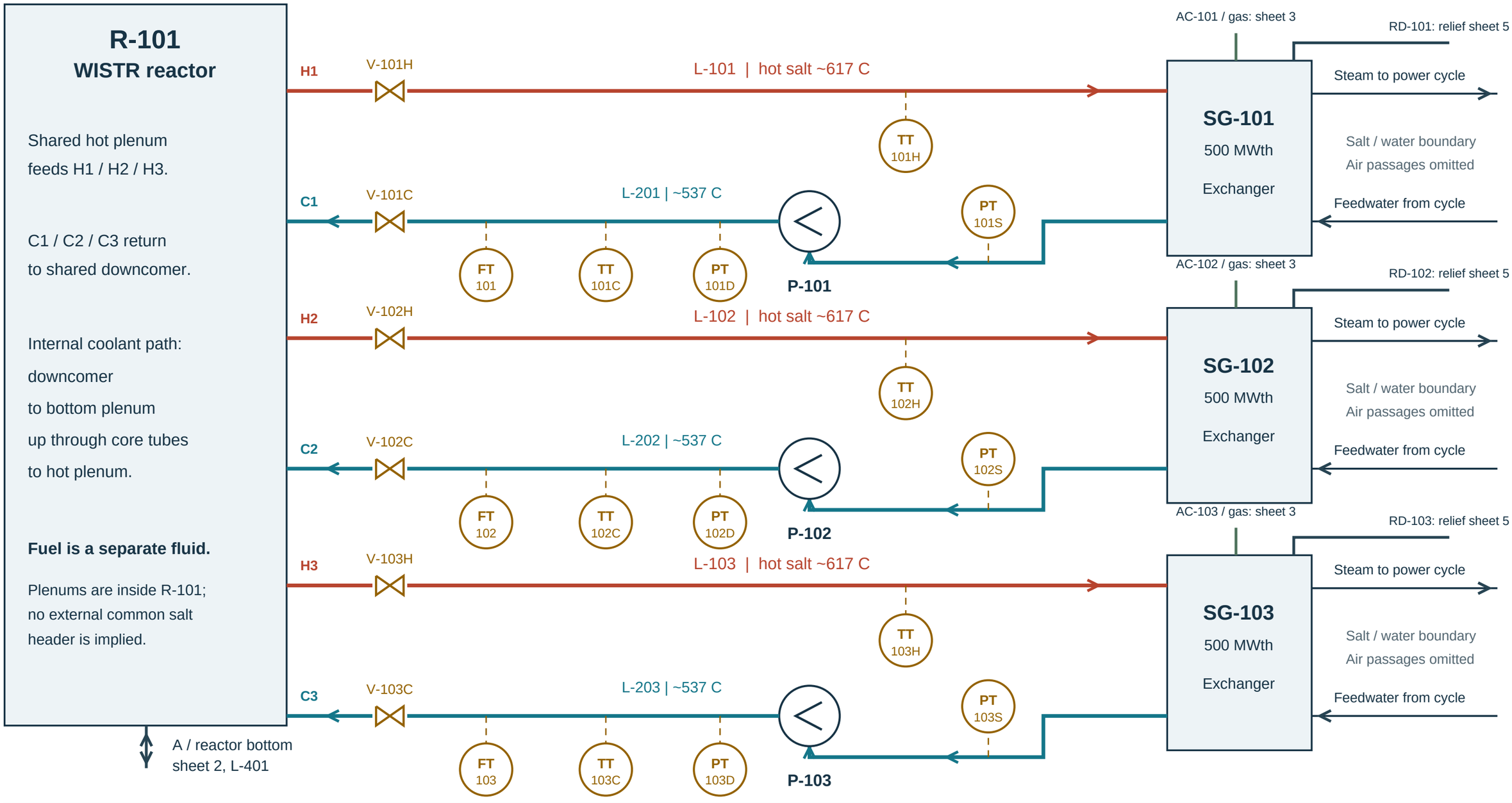
Coolant circulation

Three established process legs; proposed measurement and isolation locations shown in amber.

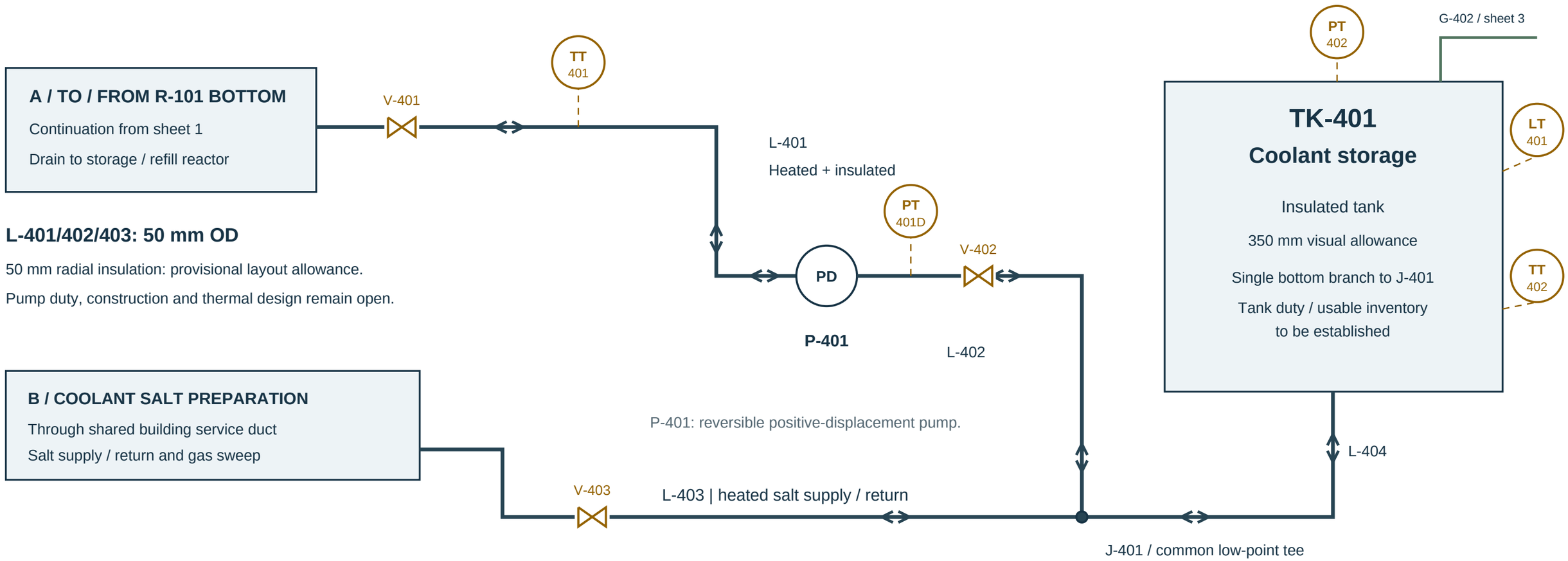


PER LEG: 3.723 m3/s; pump rise 172 kPa; 0.80 MW shaft / 0.83 MW electric; 2 MW motor reservation.

UM flow 18500 kg/s total; core 3.4 m. Itemized provisional losses; no 50% multiplier. Pump rise is not design pressure.

Coolant storage, transfer and review points

Common low-point tee; pressure gas empties TK-401 and then sweeps L-403 back to coolant preparation.



L-401/402/403: 50 mm OD

50 mm radial insulation: provisional layout allowance.

Pump duty, construction and thermal design remain open.

P-401: reversible positive-displacement pump.

DRAWING CONVENTIONS

Solid pipes: established process connections; arrows show intended flow.
Red / blue = hot / return salt; dark = service salt; green = coolant gas.
Amber valves and instrument bubbles: proposals for review.
Dashed lines: measurement association, not impulse or signal lines.

TAG KEY / PROPOSAL LIMITS

TT temperature; PT pressure; FT flow; LT level (measurement intent).
V = isolation valve; type, actuation and fail position unselected.
R reactor; SG steam generator; P pump; TK storage; L process line.
Pressure-control intent: sheet 3; settings and protection unresolved.

OPEN: OPERATING AND MAINTENANCE STATES

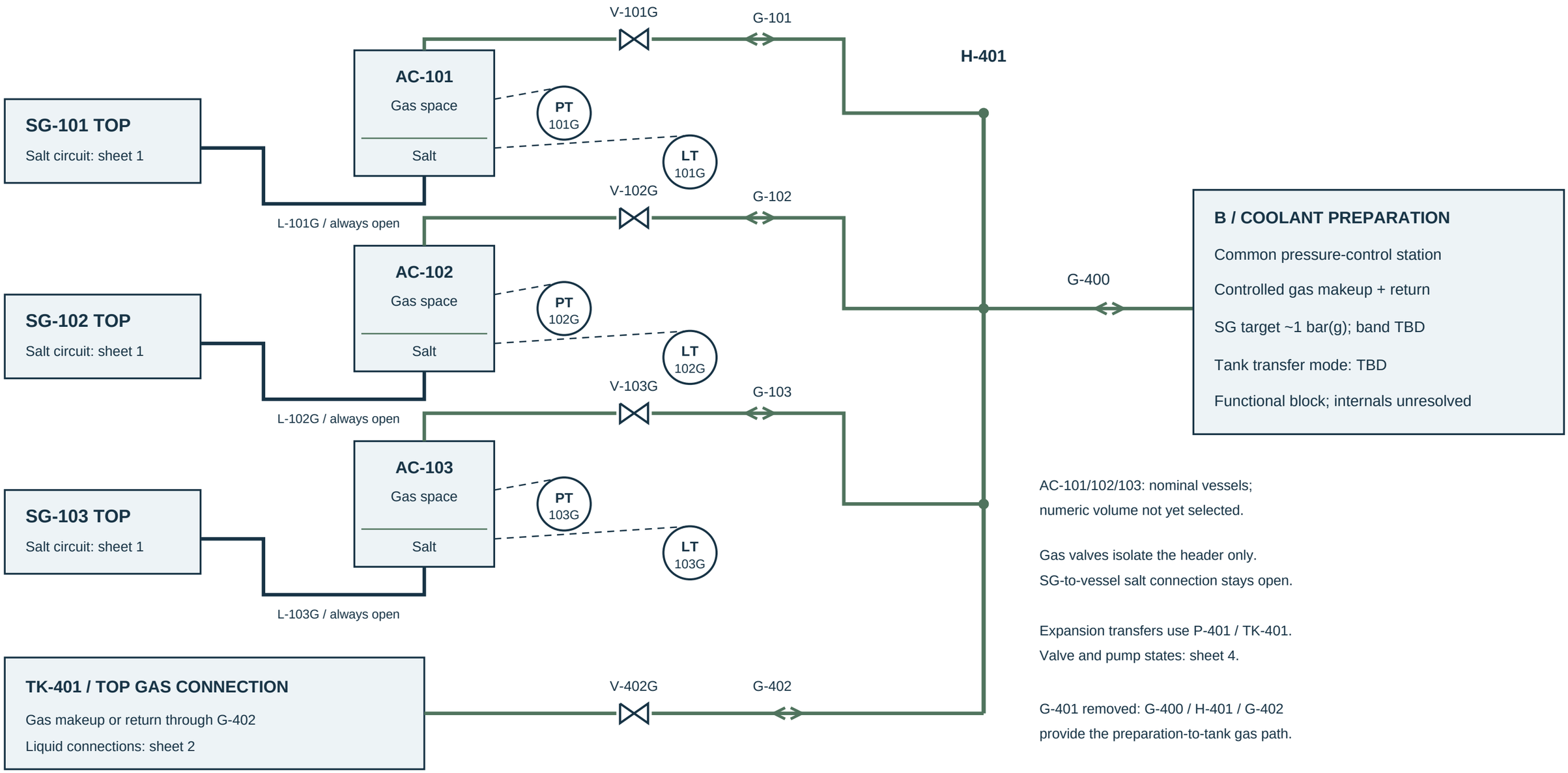
Reactor / SG drainage via P-401; tank / L-403 blowdown via G-402.
Establish inventory, tank freeboard, gas space and expansion provision.
Coolant-gas connections: sheet 3; agreed states: sheet 4.
Check pump suction conditions and shared-plenum flow distribution.

OPEN: PRESSURE BOUNDARIES AND PROTECTION

Establish pressure profile; 1 MPa(g) is only a wall-screening case.
Steam leaks to SG air side by design; resolve air-side response.
Define vent / relief destinations and trapped-salt thermal protection.
Review proposed isolation, measurement technology and remote access.

Coolant gas collection and pressure control

One common pressure-control station at preparation; coolant gas is separate from fuel off-gas treatment.



AC-101/102/103: nominal vessels;
numeric volume not yet selected.

Gas valves isolate the header only.
SG-to-vessel salt connection stays open.

Expansion transfers use P-401 / TK-401.
Valve and pump states: sheet 4.

G-401 removed: G-400 / H-401 / G-402
provide the preparation-to-tank gas path.

Clean N2 makeup: N-603, sheet 6.
Return stays separate from N2 supply.

OPEN DESIGN ITEMS

TK-401 normally nearly empty; P-401 adjusts SG vessel levels. Pressure band and transfer triggers remain unresolved.

Relief return to TK-401 is a proposal; existing closed transfer valves prevent crediting that path for relief.

Coolant operating-state draft

Agreed flow-path intent, not an operating procedure. Unassigned entries and automatic actions remain unresolved.

Operation	V-401	V-402	V-403	SG gas valves V-101G / 102G / 103G	V-402G	P-401
Normal / between transfers	closed	closed	closed	open / open / open	closed	stopped
Expansion to storage	open	open	closed	open / open / open	open	reactor to storage
Contraction from storage	open	open	closed	open / open / open	open	storage to reactor
Reactor / SG drain	open	open	closed	open / open / open	open	reactor to storage
Reactor / SG refill	open	open	closed	open / open / open	open	storage to reactor
Tank to preparation	closed	closed	open	closed / closed / closed	open	stopped
Gas sweep to preparation	closed	closed	open	closed / closed / closed	open	stopped

EXPANSION / CONTRACTION AS NEEDED

Transfer expanded salt to TK-401; return salt as the circuit contracts.

Most expansion is assigned to storage, not the nominal SG gas vessels.

Between transfers: stop P-401; close V-401, V-402 and V-402G.

Transfer triggers, level limits and usable storage volume remain TBD.

PRESSURE AND GAS PATHS

Normal circulation: SG gas valves open to the common regulated header.

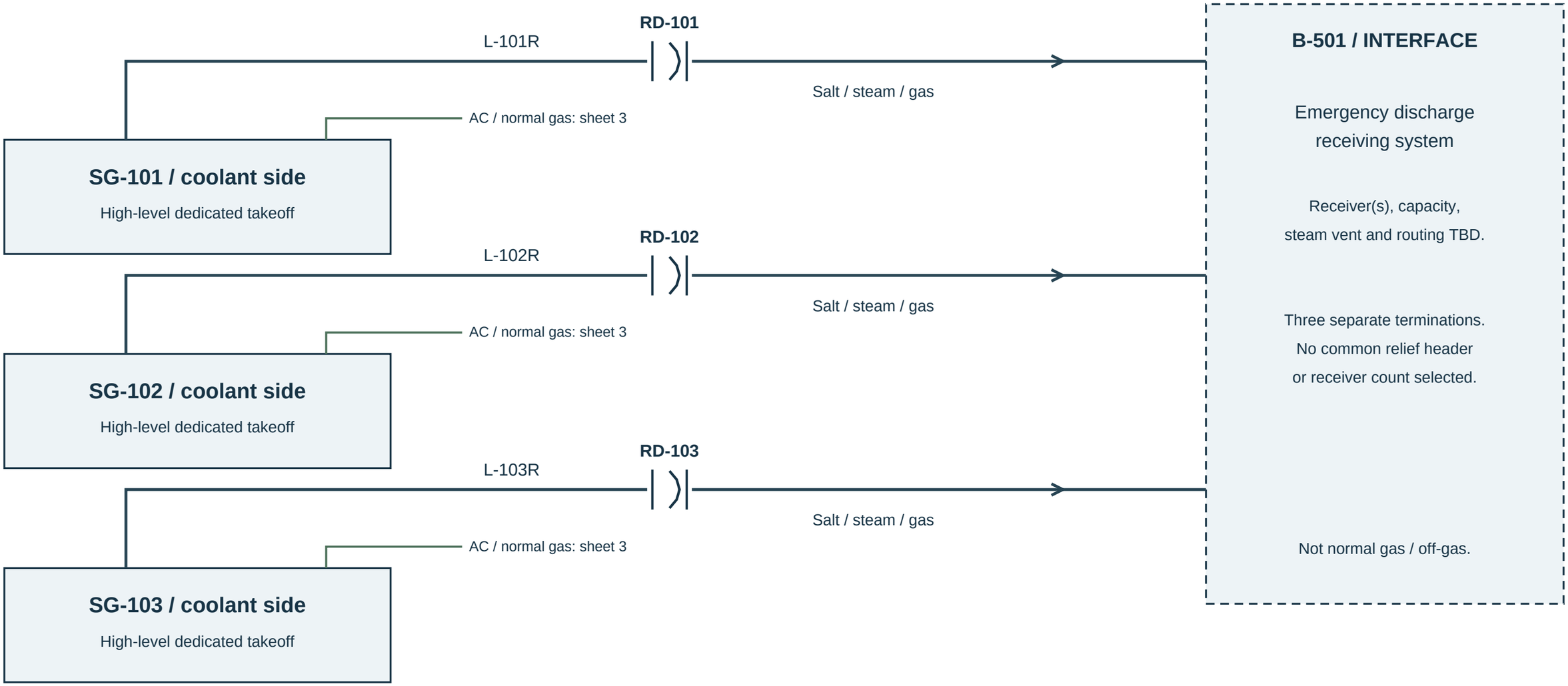
Tank emptying / sweep: isolate SG gas branches; open the TK gas branch.

Preparation > G-400 > H-401 > G-402 > TK-401 > L-404 > J-401 > L-403 > preparation.

This sweep covers the tank / return line; reactor and SG draining use P-401.

Dedicated high-level SG relief

D-062: accepted connection concept; rupture-disc sizing and discharge receiving arrangements remain unresolved.



REQUIREMENT: preserve reactor coolant inventory; prevent continued gravity drainage and siphoning after relief.

Discs normally intact; remain open after actuation. No intervening isolation valve is included in this concept.

Independent of AC-101/102/103, G-101/102/103 and H-401. Closing a normal gas valve does not isolate these paths.

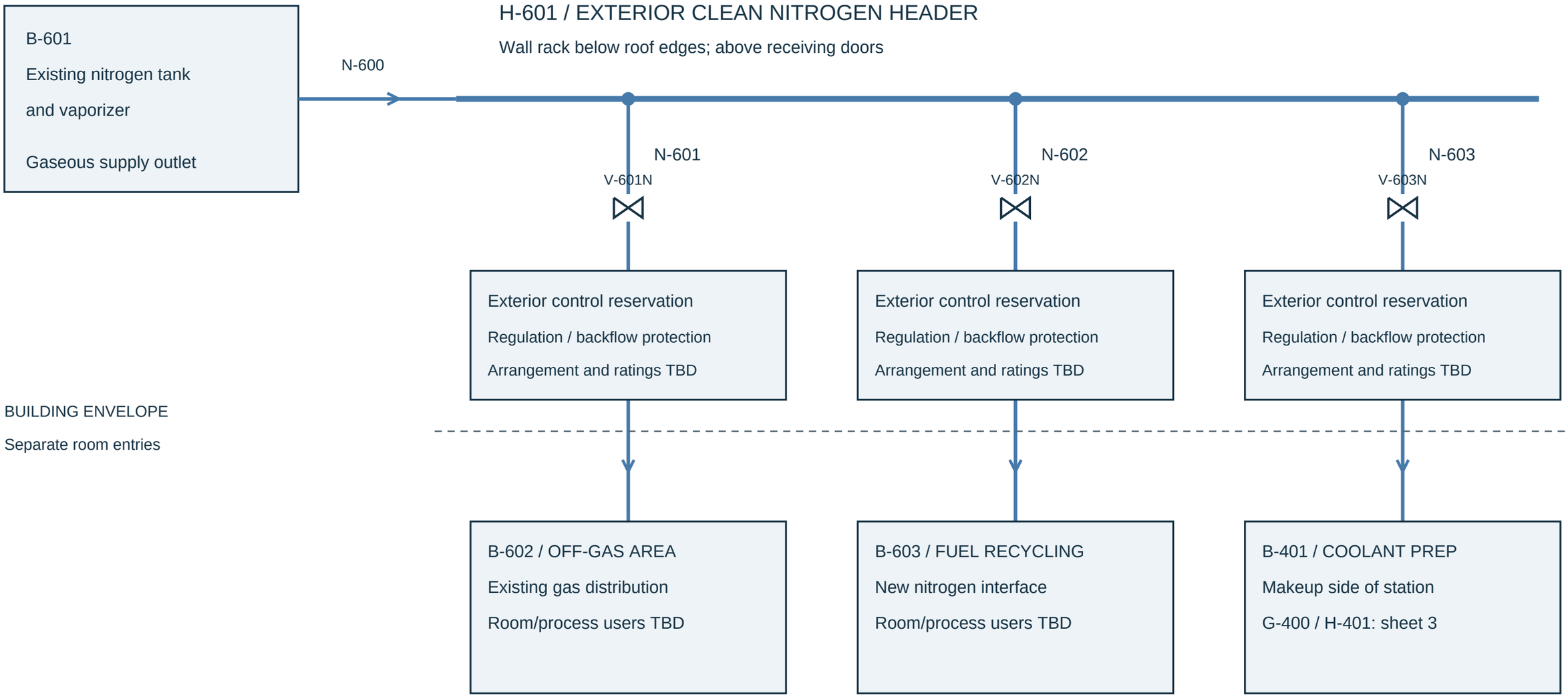
High takeoffs do not prove retention during pressure-driven discharge; expelled salt and two-phase flow remain to assess.

TBD: elevations, burst pressure at disc temperature, area/capacity, backpressure, freeze protection and post-event cooling.

Symbol: curved diaphragm between holder faces = rupture disc. Functional schematic; no physical pipe slopes implied.

Nitrogen supply - three service areas

Exterior utility header; separate room branches. Functional concept only: sizes, pressures and protection hardware TBD.



Clean supply only. Returning coolant gas and fuel off-gas do not enter H-601.

Branch blocks show required functions, not selected regulators or a credited contamination barrier.

Detailed gas demand, return handling, controls, pressure protection and penetration seals remain unresolved.